Network and System Setup Instructions

- **Network Connection:** Upon entering, first check the network settings. You need to connect to your own WiFi, specifically **"connection 1"** (the one starting with **"90"**).
- **Configuration Update:** `nano .bashrc`

  After modifying and saving the `.bashrc` file, you **must** open a new terminal window for the changes to take effect.

  > **Tip:** Alternatively, you can run `source ~/.bashrc` to apply the changes to your current session.

- **Monitor Mode:** `sudo airmon-ng start wlan0`

  Password: `xxxxx`

  — This command sets the wireless interface (antenna) to **monitor mode** to allow for packet capturing.

```
wheeltec@wheeltec:~$ sudo airmon-ng start wlan0
[sudo] password for wheeltec:

Found 5 processes that could cause trouble.
If airodump-ng, aireplay-ng or airtun-ng stops working after
a short period of time, you may want to run 'airmon-ng check kill'

  PID Name
 5014 avahi-daemon
 5327 avahi-daemon
 5479 wpa_supplicant
 5485 NetworkManager
 7075 dhclient

PHY     Interface       Driver          Chipset

phy0    wlan0mon        ath9k_htc       Atheros Communications, Inc. AR9271 802.11n
phy1    wlan1           ??????          Realtek Semiconductor Corp.
```

Execution and Environment Setup

- **Activate Environment:** `conda activate wheeltec`

  First, activate the specific Conda environment required for the project.

- **Set Monitoring Channel:** `sudo iwconfig wlan0mon channel 3`

  Configure the interface to **Channel 3** for monitoring.

  > **Note:** This step must be performed while the Conda environment is active.

- **Source Directory:** `/wheeltec_robot/src`

  Navigate to this directory to find `caction.py` and `ccfs.py`, which are used for **receiving data**.

- **Running Scripts:** When executing the code, you **must** use `sudo python3 [filename]`.

  Example: `sudo python3 caction.py`

```
th1 = th.Thread(target=cdw)
th2 = th.Thread(target=filt)
th3 = th.Thread(target=SendDataBase)
```

```
其中：cdw    为抓包函数
filt 数据获取
SendDataBase  将从上步获取的数据存入数据库
```

Navigation and Coordinate Tracking
Start Navigation: First, initiate the navigation system.

Command: python receive_tf.py

Run this script to receive the TF (Transform) coordinates.

```
rospy.init_node("tf")  # 这行代码的作用是初始化一个名为 "tf" 的 ROS 节点。这个节点可以发布消息到话题，订阅话题，提供服务，或者调用其他节点提供的服务。
lestener = tf.TransformListener()  # 创建一个监听器，监听和获取这些坐标变换信息
(trans, rot) = listener.lookupTransform("map", "base_link", rospy.Time(0))  # 获取机器人在地图坐标系中的位置和方向，其中 "map" "base_link" 是两种不同的坐标系
goal_status_sub = rospy.Subscriber("/move_base/result", MoveBaseActionResult, callback, queue_size=1)  # 创建一个订阅者来订阅"/move_base/result"话题。这个话题通常用于发布机器人导航的结果。
```

Publishing Position Coordinates

- **Script:** `issue_position.py`
- **Action: Publish Position** Run this script to broadcast or publish the current position/coordinate data to the system.

Communication between the two boards of the car

```
两块板子连接 ：https://blog.csdn.net/Wild_Ray/article/details/115311103


echo 'echo 1 > /proc/sys/net/ipv4/ip_forward' | sudo tee -a /etc/rc.local
echo 'iptables -t nat -A POSTROUTING -o usb0 -j MASQUERADE' | sudo tee -a /etc/rc.local
echo 'iptables -A FORWARD -i eth0 -o usb0 -m state --state RELATED,ESTABLISHED -j ACCEPT' | sudo tee -a
/etc/rc.local
echo 'iptables -A FORWARD -i usb0 -o eth0 -j ACCEPT' | sudo tee -a /etc/rc.local
sudo vim /etc/rc.local
在第一行添加上：#!/bin/sh -e
sudo chmod +x /etc/rc.local
sudo systemctl start rc-local.service
```

Navigation Logic

```
开启ros导航： roslaunch turn_on_wheeltec_robot navigation.launch
地图更换在navigation.launch中
打开rviz软件：rviz   ——进行实时监控
打开tf坐标接收：cd  ……     python receive_tf.py
发布坐标：issue_position.py x y a b 其中x, y 为位置坐标：a, b 为方向控制
前 0 1
后 -1 0
左 1 1
后 1 -1
issue_position.py：巡航逻辑是按照点的循环进行
```

Graph construction logic

```
开启ros建图：roslaunch turn_on_wheeltec_robot mapping.launch
打开rviz软件
打开键盘控制：roslaunch wheeltec_robot_rc keyboard_teleop.launch
b键可以切换运动逻辑
保存：
打开地图路径：
cd /home/wheeltec/wheeltec_robot/src/turn_on_wheeltec_robot/map
手动保存地图：
rosrun map_server map_saver -f 20220615
```

Packet receiving logic

```
开启wlan1监听模式： sudo airmon-ng start wlan1
                 sudo iwconfig wlan1mon channel 3
                 sudo python3 M_main.py
```

Car movement calibration

```
实际前进1m，里程计x是否是+1m，误差±0.02m正常
实际左移1m，里程计y是否是+1m，误差±0.02m正常
实际逆时针旋转1圈，里程计z是否是+6.28rad，误差±0.2rad正常

里程计查看命令：rostopic echo /odom
话题里面的position信息就是里程计

假设实际前进1m，里程计显示x是1.2，那么x_scale参数应该从默认的1改为1/1.2=0.833，y轴、z轴同理
下面的positive是指左转时候的误差，negative是指右转的时候的误差
```

- kcf_track
- ldlidar_14
- ldlidar_19
- lsn10p
- lsx10
- navigation-melodic
- qt_ros_test
- realsense-ros-development
- robot_pose_ekf
- ros_astra_camera
- ros_tensorflow
- rplidar_ros
- rrt_exploration
- simple_follower
- slam_karto
- teb_local_planner-melodic-devel
- tts_make
- turn_on_wheeltec_robot
  - .fang_cloud_tmp_dir
  - include
  - launch
    - .fang_cloud_tmp_dir
    - include
      - .fang_cloud_tmp_dir
      - algorithm_gmapping.launch
      - algorithm_hector.launch
      - algorithm_karto.launch
      - amcl.launch
      - base_serial.launch
      - dwa_local_planner.launch
      - dwa_local_planner_pure3d.launch
      - robot_pose_ekf.launch
      - rtabmap_mapping.launch
      - rtabmap_nav.launch
      - teb_local_planner.launch
      - teb_local_planner_pure3d.launch
      - velocity_smoother.launch
    - 3d_mapping.launch
    - 3d_navigation.launch
    - ar_label.launch

```
  <!--打开节点wheeltec_robot，初始化串口等操作-->
  <arg name="smoother"  default="false"/> <!-- 是否开启速度平滑功能 -->
  <arg name="odom_frame_id"  default="odom_combined"/> <!-- 设置里程计话题起点TF坐标 -->
  <arg name="use_FDI_IMU_GNSS"  default="false"/> <!-- 是否使用飞迪系的惯导、GNSS/RTK模块，默认false

  <node pkg="turn_on_wheeltec_robot" type="wheeltec_robot_node" name="wheeltec_robot" output="scr
    <param name="usart_port_name"    type="string" value="/dev/wheeltec_controller"/>
    <param name="serial_baud_rate"   type="int"    value="115200"/>
    <param name="odom_frame_id"      type="string" value="$(arg odom_frame_id)"/>
    <param name="robot_frame_id"     type="string" value="base_footprint"/>
    <param name="gyro_frame_id"      type="string" value="gyro_link"/>

    <!-- 里程计修正参数，对来自运动底盘的速度信息进行手动修正 -->
    <param name="odom_x_scale"      type="double" value="1.0"/>
    <param name="odom_y_scale"      type="double" value="1.0"/>
    <param name="odom_z_scale_positive"      type="double" value="1.0"/>
    <param name="odom_z_scale_negative"      type="double" value="1.0"/>

    <!-- 如果开启了平滑功能，则订阅平滑速度 -->
    <remap if="$(arg smoother)" from="cmd_vel" to="smoother_cmd_vel"/>

    <!-- 如果使用飞迪系的惯导、GNSS/RTK模块，对板载IMU话题进行重命名，避免相关节点订阅板载IMU话题 -->
    <remap if="$(arg use_FDI_IMU_GNSS)" from="imu" to="imu_on_board"/>
  </node>

  <!-- 如果使用飞迪系的惯导、GNSS/RTK模块，启动相关launch文件，该节点发布的IMU话题名称就是【/imu】 -->
  <include if="$(arg use_FDI_IMU_GNSS)"
    file="$(find fdilink_ahrs)/launch/ahrs_data.launch" >
  </include>

  <!-- 如果开启了速度平滑功能，则运行速度平滑功能包 -->
  <include if="$(arg smoother)"
    file="$(find turn_on_wheeltec_robot)/launch/include/velocity_smoother.launch" >
  </include>

</launch>
```